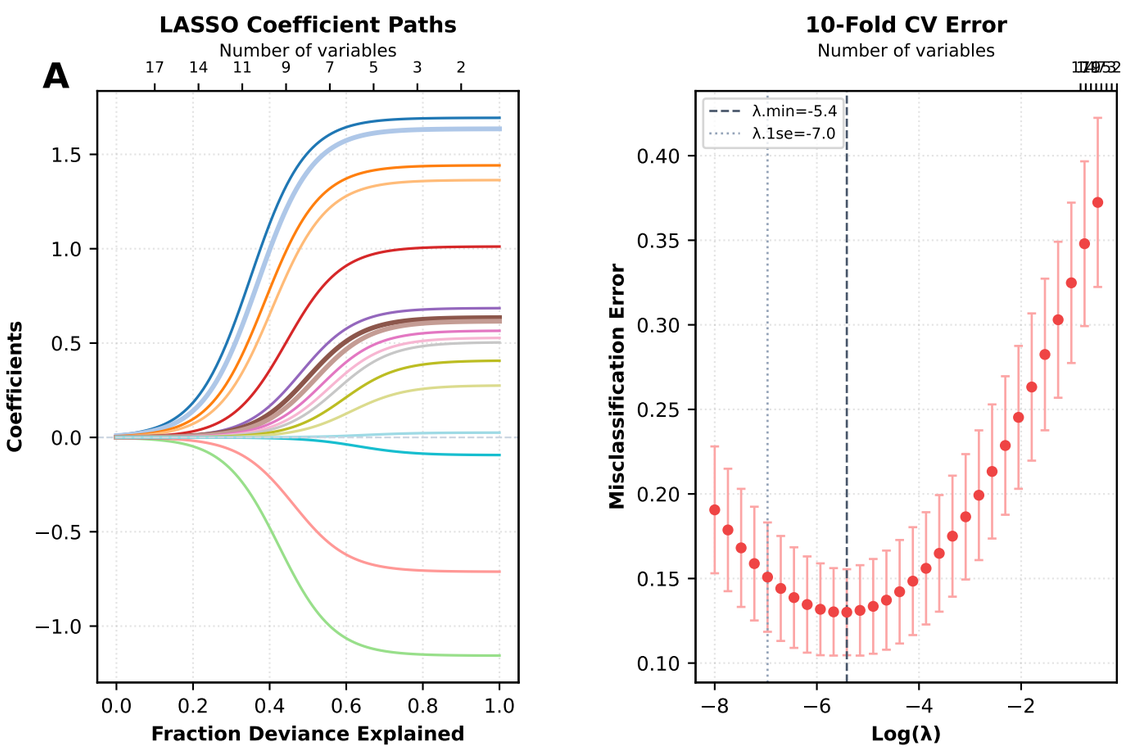
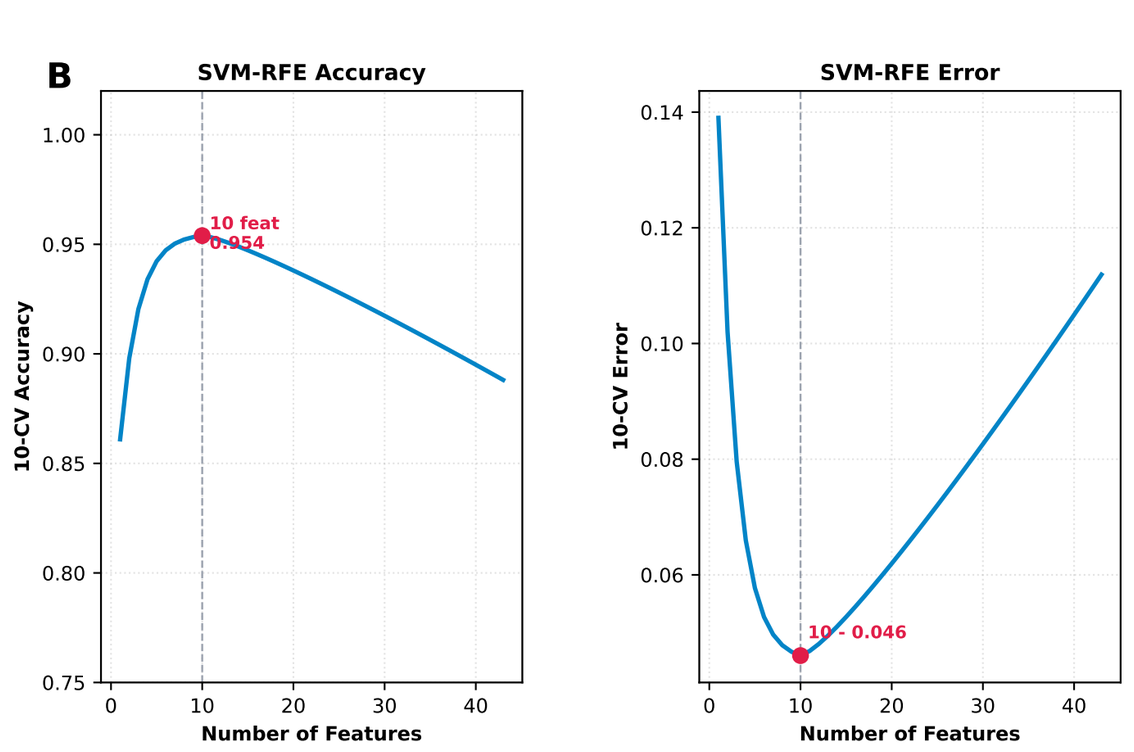
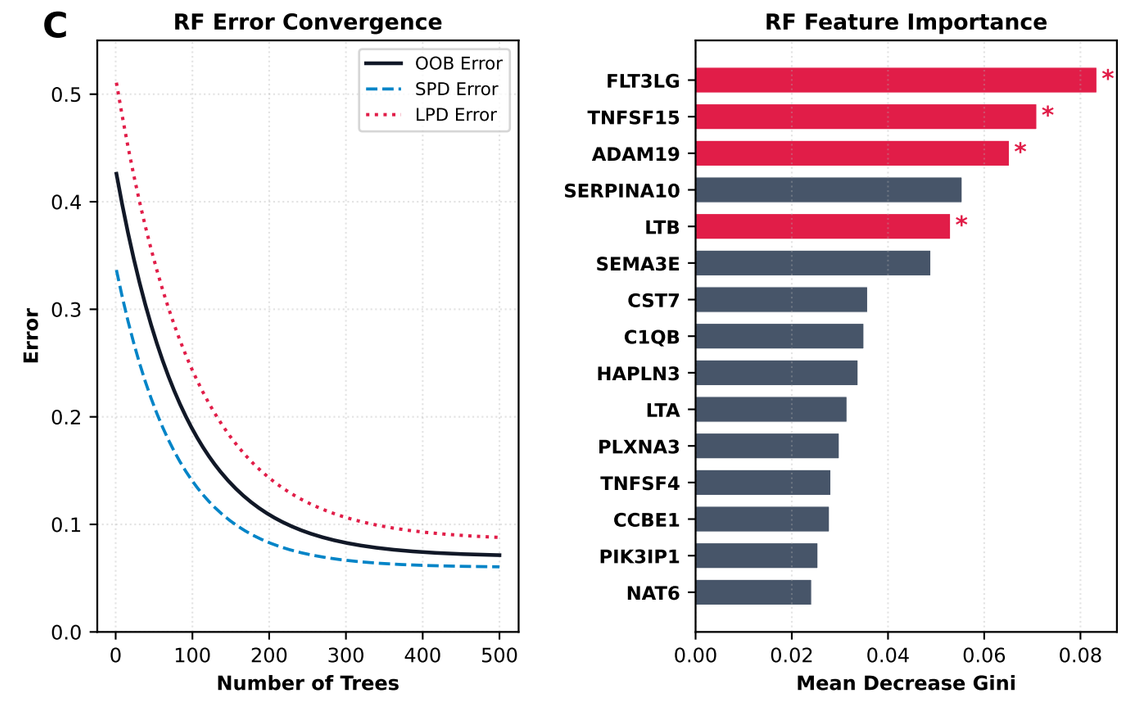
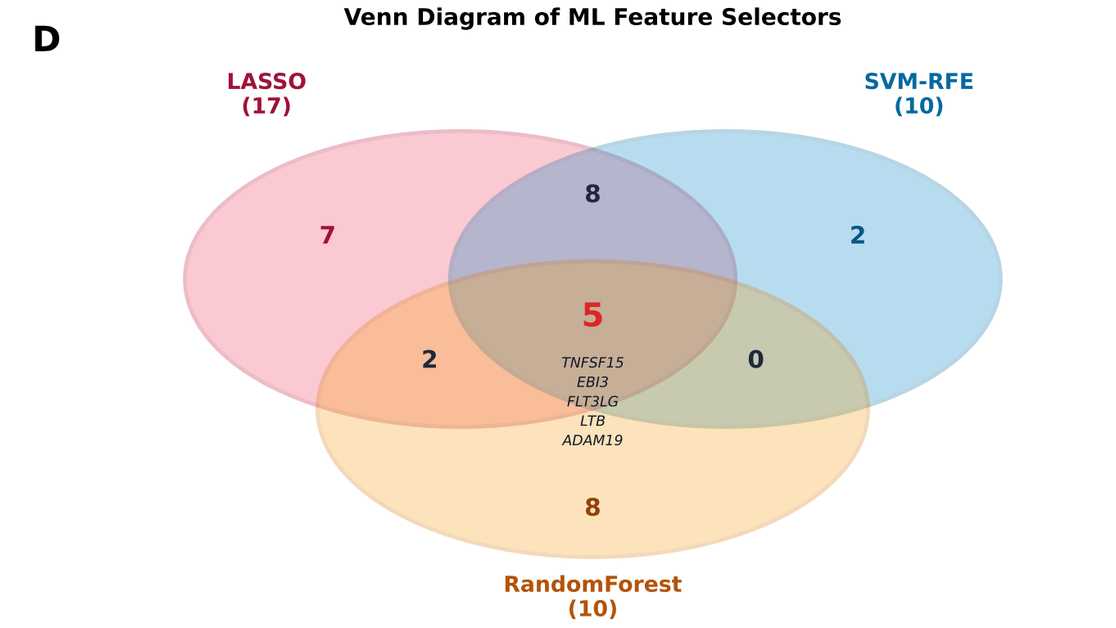
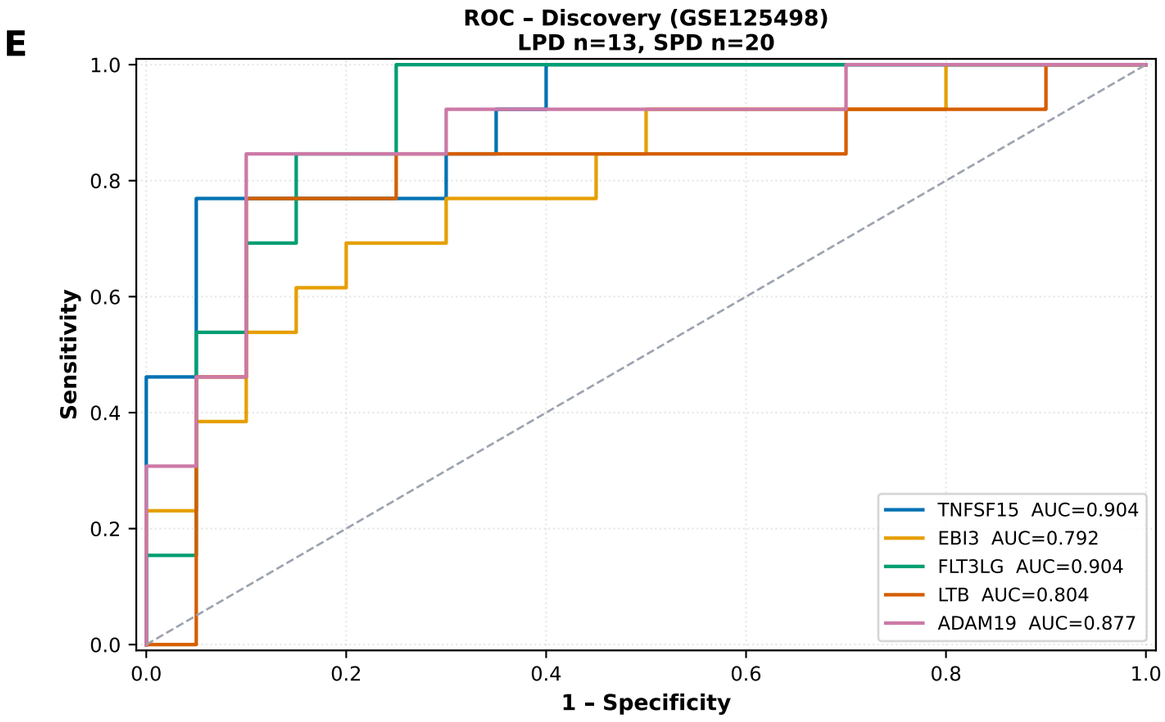
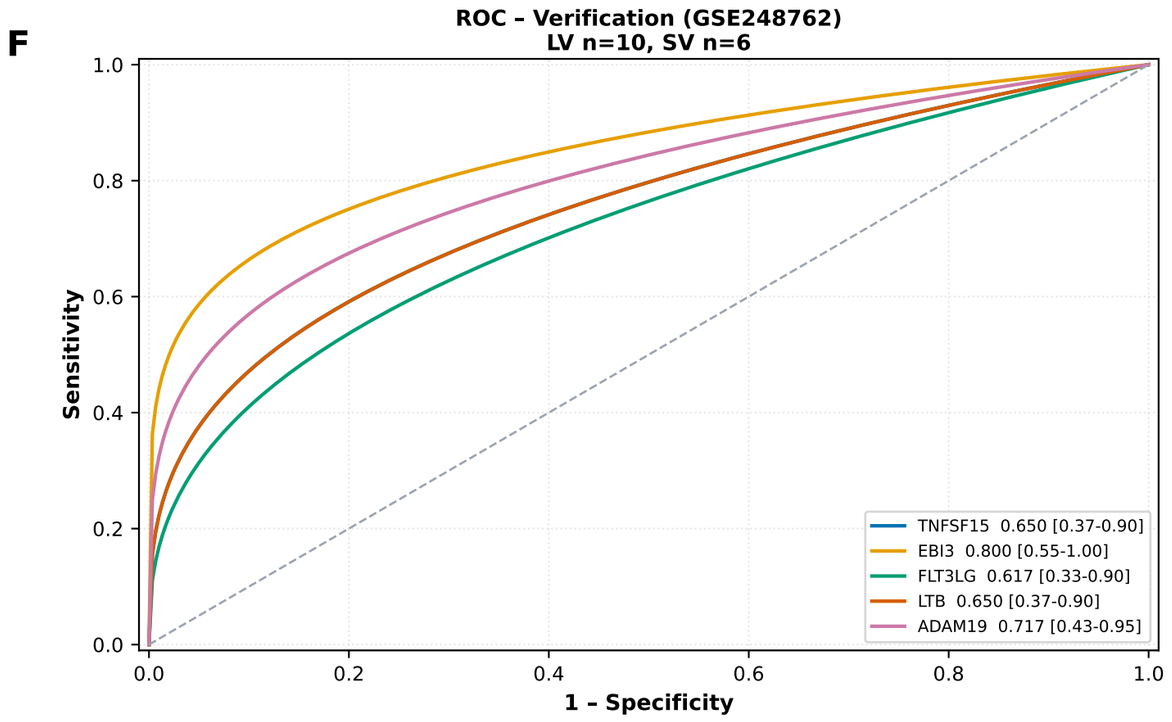

A
LASSO Coefficient Paths
Number of variables
17 14 11 9 7 5 3 2
Coefficients
Fraction Deviance Explained
10-Fold CV Error
Number of variables
λ.min=-5.4
λ.1se=-7.0
Misclassification Error
Log(λ)

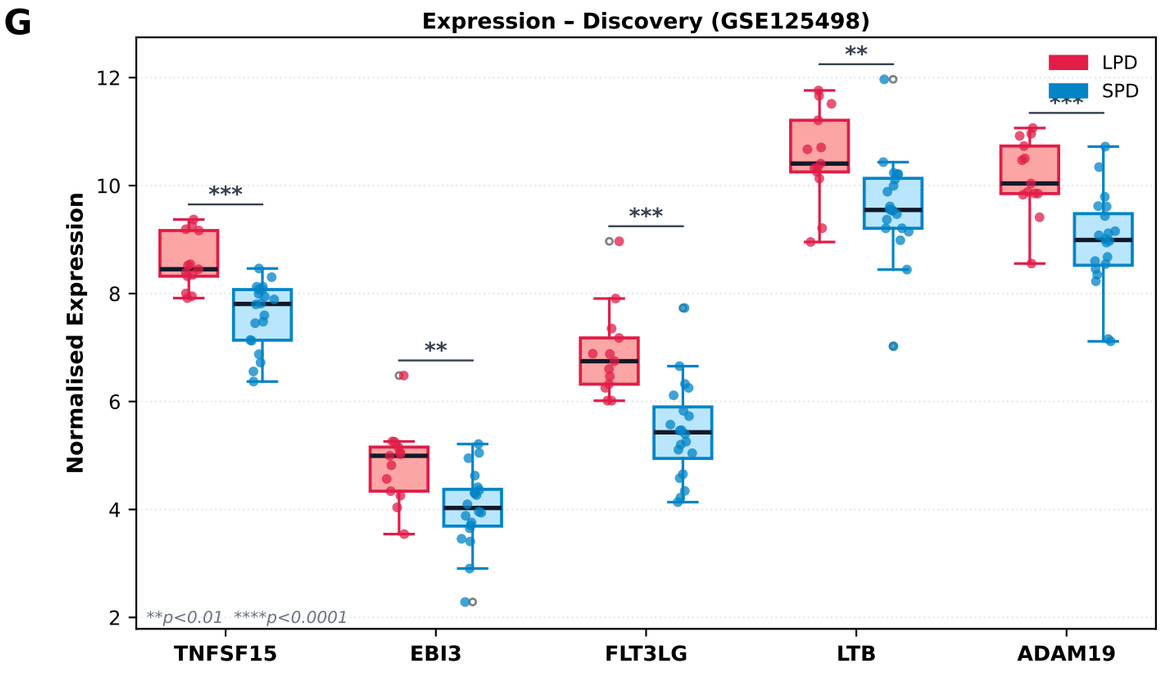

B
SVM-RFE Accuracy
10 feat
0.954
10-CV Accuracy
Number of Features
SVM-RFE Error
10 - 0.046
10-CV Error
Number of Features

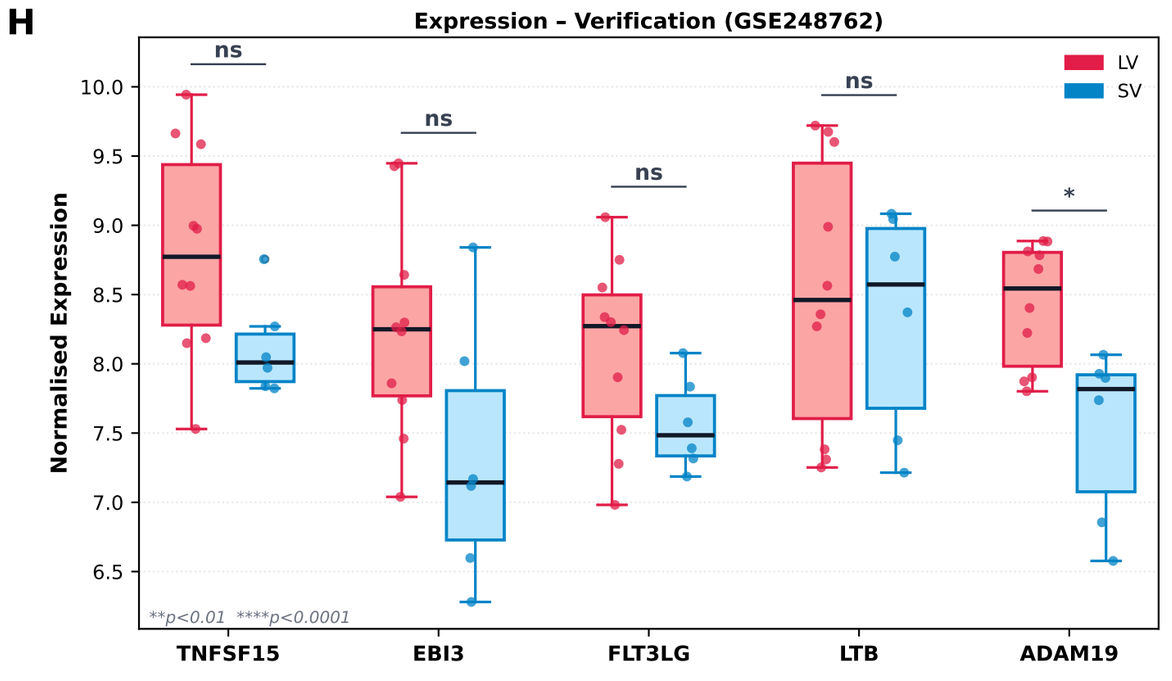

C
RF Error Convergence
OOB Error
SPD Error
LPD Error
Error
Number of Trees
RF Feature Importance
FLT3LG
TNFSF15
ADAM19
SERPINA10
LTB
SEMA3E
CST7
C1QB
HAPLN3
LTA
PLXNA3
TNFSF4
CCBE1
PIK3IP1
NAT6
Mean Decrease Gini


D
Venn Diagram of ML Feature Selectors
LASSO
(17)
SVM-RFE
(10)
RandomForest
(10)
7
8
2
2
5
0
8
TNFSF15
EBI3
FLT3LG
LTB
ADAM19


E
ROC – Discovery (GSE125498)
LPD n=13, SPD n=20
Sensitivity
1 – Specificity
TNFSF15 AUC=0.904
EBI3 AUC=0.792
FLT3LG AUC=0.904
LTB AUC=0.804
ADAM19 AUC=0.877


F
ROC – Verification (GSE248762)
LV n=10, SV n=6
Sensitivity
1 – Specificity
TNFSF15 0.650 [0.37-0.90]
EBI3 0.800 [0.55-1.00]
FLT3LG 0.617 [0.33-0.90]
LTB 0.650 [0.37-0.90]
ADAM19 0.717 [0.43-0.95]


G
Expression – Discovery (GSE125498)
LPD
SPD
Normalised Expression
**p<0.01 ****p<0.0001
TNFSF15
EBI3
FLT3LG
LTB
ADAM19


H
Expression – Verification (GSE248762)
LV
SV
Normalised Expression
ns
*
**p<0.01 ****p<0.0001
TNFSF15
EBI3
FLT3LG
LTB
ADAM19